Verity — hackFront India 2026 Build Plan

Oct 26, 2026 · @Ansh

1. Start here

Team Low Cortisol is building **Verity** for hackFront India 2026: a trip planner that tells an electric-vehicle (EV) driver, with an honest confidence level, whether they will reach their destination, and what to do the moment a charger fails. The prototype is due Oct 31, 2026, 11:59 PM IST.

Item	Detail
Competition	hackFront India 2026, a nationwide innovation league run by Where U Elevate
Team	Low Cortisol (team lead: Ansh Goyal)
Theme	T6 Automobile and EV (Deep Tech and Industry sector)
Problem statement	T6.1 "Will I Make It": build a routing and energy prediction engine that tells an Indian driver honestly whether they will make it, and plans charging around chargers that actually work
Partner tracks we enter	Neo4j (charger graph for risk-aware planning) and Sarvam (voice alerts in the driver's own language)
What we submit	A one-line problem statement and a prototype deck (PDF or PPTX), plus a working demo
Prototype deadline	Oct 31, 2026, 11:59 PM IST, online
Last day to add teammates	Oct 27, 2026, 11:59 PM IST (registration closes)
National finale	Nov 21, 2026 to 22 Nov 2026: 22-hour hackathon in person at IIIT Delhi
After the finale	Go-to-market cohorts and hiring mapping, 1–27 Dec 2026

Item	Detail
Prizes	Overall ₹1,00,000 / ₹50,000 / ₹25,000 · Neo4j track ₹25,000 / ₹15,000 / ₹10,000 plus credits · Sarvam track API credits

How to read this doc: sections 2–4 explain the idea and why it wins, section 5 shows how it works, sections 6–9 are this week's plan, and sections 10–12 are reference. Section 12 defines every technical term.

2. The idea in one page

Verity treats every EV trip the way an airline treats a flight: as a dispatch decision with a probability, a backup plan and a hard "divert now" threshold, instead of one range number.

The insight

Today an EV driver sees one number, such as "Range 230 km". It comes from a gentle lab test, the industry's own advice is to knock 20–30% off it, and the chargers on the map may be dead. Airlines never dispatch a flight on one fuel number. They carry fuel for the destination, a reachable alternate airport and a reserve, and pilots have a "bingo fuel" level at which they must divert. Verity gives every EV trip the same discipline.

The driver asks	A typical app today	Verity
Will I make it?	"You'll arrive with 19%"	"About 90% confident; you'll arrive with 8–19%"
What if the charger is dead?	Finds out on arrival	A reachable alternate is planned from every kilometre
When must I act?	Never told	The **Bingo Line**: the battery % below which you must divert, shown for every kilometre
Why this estimate?	Not shown	A breakdown, e.g. AC −11%, climb −6%, four passengers −5%
Can I improve my odds?	No	Levers, e.g. "cruise at 90 instead of 100" or "charge 9 more minutes at stop 1"
In what language?	English text	Voice alerts in the driver's own language

Numbers in this table are illustrative; the real ones come from the model.

Who it is for

- **The driver** gets the honest answer on their phone, by voice, while driving.
- **The fleet operator** pays. Their question is "can vehicle 12 take this Agra booking, and where must it charge?" A dead vehicle in the wrong place costs them a whole shift.
- **Used-EV owners and buyers** benefit too: Verity learns a vehicle's real usable battery capacity from a few trips, which nobody in India certifies today.

Four things a default solution will not have

1. **Plans with a backup, not one route.** Every plan includes alternates, and the success probability accounts for diverting when a charger fails.
2. **No false confidence on long trips.** Errors that persist through a whole trip (driver style, heat, load, battery age) are modelled as shared, so long trips are not wrongly treated as safer.
3. **A charger trust graph.** It combines crowd reports, operator status feeds and usage traces, and learns which sources lie.
4. **Useful after two or three trips.** A physics model plus fleet-wide priors calibrates each vehicle from very little data.

The abstract we will submit

Title: Verity: Dispatch-Grade Honesty for Every EV Trip

One-line problem statement (form field): Indian EV drivers and fleets get one optimistic range number and a map of chargers that often don't work; Verity tells them, with a tested confidence level, whether they'll make it, and what to do the moment a charger fails.

> India's EV drivers plan trips on a single number, a claimed range that heat, AC and highway speed routinely cut by a fifth or more, across a charging network where audits keep finding dead chargers. Airlines never dispatch a flight on one fuel number; they plan a destination, a reachable alternate and a reserve. Verity brings that discipline to EVs. Every trip gets a probability of arrival, a live Bingo Line (the battery level below which the driver must divert), and alternates that stay reachable from every kilometre. Underneath, a physics energy model learns each vehicle in a few trips using priors pooled across a fleet. Uncertainty shared across a trip (driver, heat, load, battery age) is modelled explicitly, so long trips are never falsely confident. A Neo4j charger graph infers reliability from noisy reports, operator feeds and usage traces, while learning which sources to trust. On trips it never saw, [X]% of outcomes fell inside our 90% band, and reliability-aware planning cut simulated strandings by [Y]%.

Fill [X] and [Y] only with measured results on Friday. Never submit invented numbers.

3. Why this problem is real

Three facts carry the pitch: real range falls well short of the claim, a large share of public chargers fail, and no existing app turns that mess into a probability. Sources were found by web search on 26 Oct 2026; re-check any figure before it goes on a slide.

Area	Finding	Why it matters for Verity	Source
Range	Autocar India's highway test of the Curvv EV 55 got 359 km; Tata claims 400–425 km	The brochure number misleads even on a good road	Autocar Professional
Range	Indian test cycle averages 31 km/h with a 90 km/h top speed	Lab cycles do not resemble highway driving	Autocar Professional
Range	Standard advice: expect at least 20–30% less than claimed	The industry's fix is a flat discount, still one number. This is our foil	Gaadify

Area	Finding	Why it matters for Verity	Source
Range	A 100 km certified cargo three-wheeler does about 70–80 km with load, AC and an aged pack	Commercial vehicles suffer most	ev.care
Range	Pune drive cycles ranged from 84 to 261 Wh/km, driven by aggressiveness	Driver style is a huge factor and must be learned	Andhale et al., SIELMEN 2025 (listing)
Chargers	A 2024 IEEFA audit found about 84% of chargers examined in Delhi non-functional (small sample)	Charger risk dominates trip risk	IEEFA briefing
Chargers	Chargers listed as "working" online were not functional on the ground	Operator status feeds cannot be trusted at face value	IEEFA briefing
Chargers	Parliament data: 27,737 stations installed, 22,753 operational (Mar 2026); BPCL's network reportedly 60% non-operational	Reliability varies sharply by network, which gives us priors	MMCM
Chargers	About 12,100 of 25,000 chargers were non-functional at some point in Feb 2024; users juggle 17–20 apps	Fragmented, unreliable data	Business Standard
Chargers	A fleet playbook puts Indian DC fast-charger downtime near 40% and says plans must assume a failed charger	Backup plans are a must, not a feature	Yo Mobility
Chargers	88% of EV owners name finding a working, accessible charger as a main anxiety	The pain is charger trust, not maps	Exicom

Area	Finding	Why it matters for Verity	Source
Chargers	Heat can slow fast charging by 10–30%	Driving hard in heat makes the next stop longer	Bolt.Earth
Chargers	One charger brand triggered a fault on a Nexon EV that blocked fast charging	Reliability depends on the car–charger pair	Team-BHP
Existing tools	ABRP users have requested warnings when no alternate charger is reachable	The best global planner lacks backup logic	ABRP feature board
Existing tools	Indian apps (Pulse, ChargeIndia, EVz, Electree) offer check-ins, live status and photos	Evidence exists; nobody turns it into a probability	Pulse, ChargeIndia
Fleets	An operator described range anxiety as usable hours in a day, not kilometres per charge	Fleets measure the problem in time and rupees	ev.care
Fleets	Longer charging directly cuts daily trips and driver earnings	Clear willingness to pay	Outlook Business
Used EVs	India has no accepted standard for certifying used-EV battery health	Our capacity estimate fills a market gap	Vahan Bazaar
Used EVs	A 2026 study of 1,114 EVs found gaps between true capacity and the car's own battery-health figure	The car's own number cannot be trusted either	Analytics Insight
Research	Energy use on different road segments is statistically dependent	Treating segments as independent makes long trips look falsely safe	EasyChair preprint

Area	Finding	Why it matters for Verity	Source
Research	Uncertainty quantification in battery-EV energy modelling remains sparse	Calibrated confidence is genuinely new ground	arXiv 2511.06538
Data	Open Charge Map says it has little India data	Charger lists must be merged and checked by hand	OCM forum
Data	TU Munich published 72 real BMW i3 trips with temperature, elevation and battery data	A real held-out test set we can use this week	IEEE DataPort

4. What the judges reward, and our proof for each

Every judging criterion has a named proof in our build, so nothing in the deck is a claim without evidence. Criteria come from three places: the general rubric on the hackathon page, the T6.1 problem statement, and the Neo4j and Sarvam track briefs.

What these organisers value

The same organisers wrote all eight themes' problem statements, and the same preferences repeat across them:

- **Real measured data beats simulation.** One statement says 200 honest measured points beat a simulated million.
- **Uncertainty comes with its sample size,** e.g. "based on 3 reports in 60 days".
- **Crowdsourced data needs a manipulation defence,** thought through rather than mentioned.
- **Refuse rather than invent** when the data does not support an answer.
- **Talk to real users** and say what you changed because of them.
- **Physics and explainability over black boxes,** and state your compute cost.
- **Win on "the layer beneath"**: the hard part under the surface. For T6.1 that is calibration from sparse data and planning under uncertainty, which the statement says is a different problem from finding a shortest path.

The scorecard

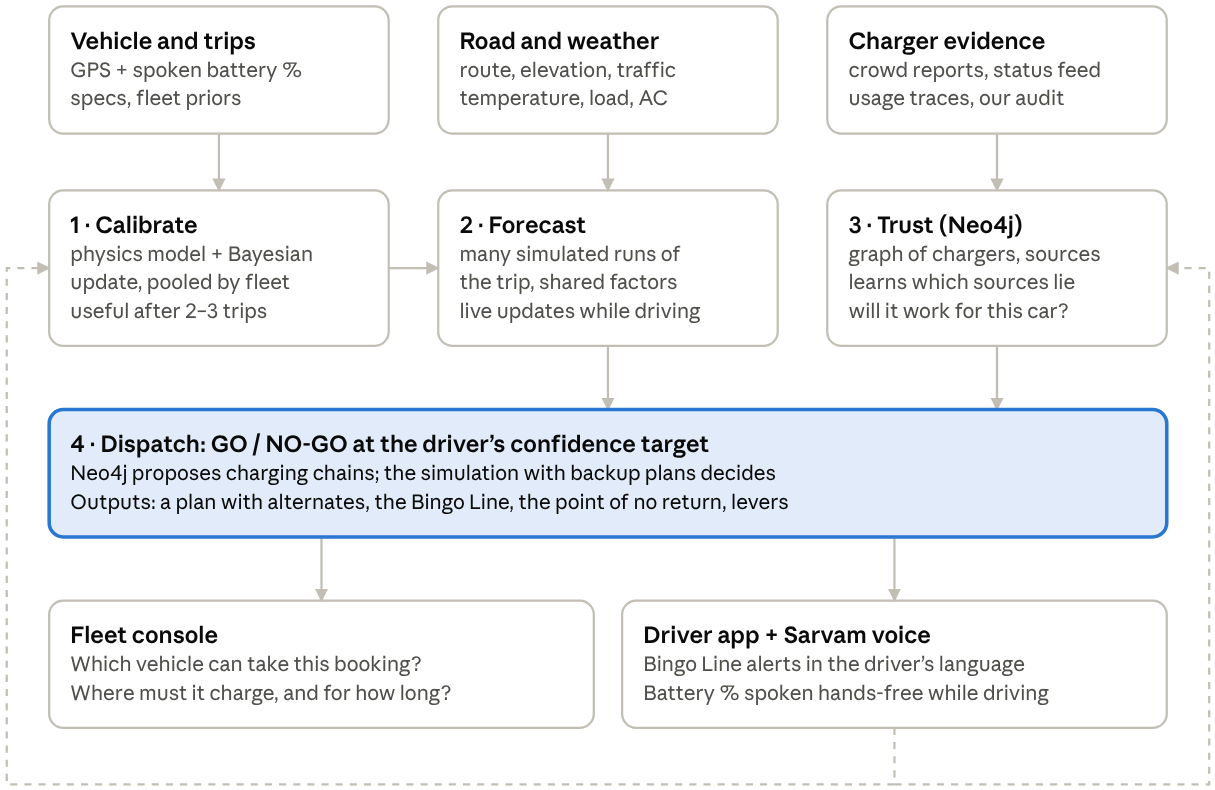
Criterion	Where it comes from	Our proof	Where judges see it
Problem and track alignment	General rubric	Every bullet of T6.1 implemented; Neo4j and Sarvam do real work	Architecture slide, mapped bullet by bullet
Innovation and effectiveness	General rubric	Dispatch with backups, Bingo Line, shared uncertainty, charger trust graph	Slides 3, 7, 9
Technical implementation and code quality	General rubric	Typed Python package; physics sanity tests; one-command experiment scripts; clean README	Repository
Functionality and validation	General rubric	Experiments E1–E5 and a live demo	Slides 6–9, demo video
Critical thinking	General rubric	Limitations slide; shortest-path vs honest-planning comparison	Slides 9 and 12
Prediction error on unseen trips	T6.1	E1: held-out error and learning curve	Slide 6
Charger reliability changes decisions	T6.1	E3: decision-change rate and strandings, plus a real audited charger	Slide 8
Honest uncertainty in the interface	T6.1	Confidence bands, sample sizes, refusal to promise a trip	Demo
Calibration from sparse data	T6.1 layer beneath	E1 learning curve: useful after 2–3 trips	Slide 6
Stochastic planning, not shortest path	T6.1 layer beneath	E4: fixed plan vs dispatch policy	Slide 9

Criterion	Where it comes from	Our proof	Where judges see it
Graph for risk-aware trip planning	Neo4j track	Charger trust graph, fake-report ring detection, candidate chains, coverage gaps	Slide 8
Voice guidance in the driver's language	Sarvam track	Spoken battery input and Bingo Line alerts	Slide 10, demo

5. How Verity works

Four engines turn raw inputs into one decision: Calibrate learns the vehicle, Forecast predicts the arrival, Trust scores the chargers, and Dispatch decides and guides the driver.

Four engines turn messy inputs into one honest decision



Dashed: each finished trip re-calibrates the vehicle; each charger report updates the graph

Verity architecture · 3 inputs, 4 engines, 2 interfaces

Solid arrows are the planning path from inputs to a decision; dashed arrows are the learning loop that makes every trip improve the next one.

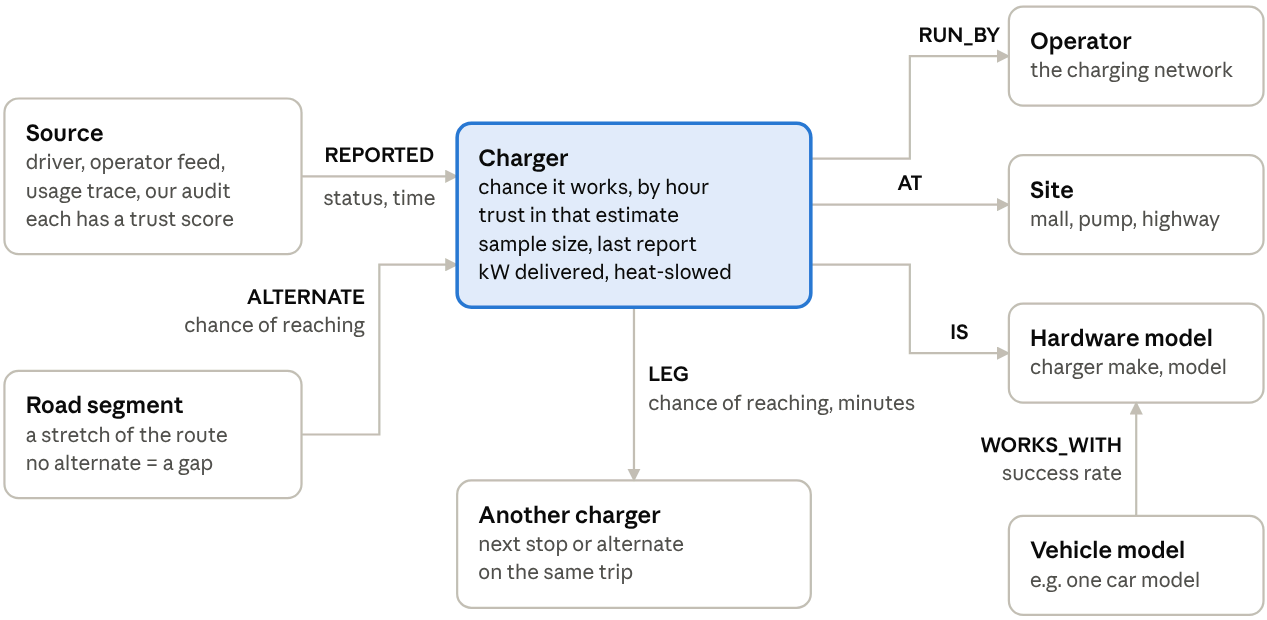
The four engines

Engine	Job	Inputs	Output	Key method
Calibrate	Learn each vehicle	Trip logs (GPS plus battery %), vehicle specs, fleet priors	The vehicle's efficiency, AC load, usable capacity and driver factor, each with its uncertainty	Physics model plus Bayesian update, pooled across a fleet

Engine	Job	Inputs	Output	Key method
Forecast	Predict the arrival honestly	Route, elevation, traffic, weather, payload, calibrated vehicle	A spread of likely arrival battery %, not one number	Monte Carlo with shared trip-level factors; live updates during the trip
Trust	Know which chargers work	Crowd reports, operator status, usage traces, our audit, network and site type	Chance each charger works for this car at that hour, and expected minutes to get the energy needed	Neo4j graph, source-trust model, priors by network and site, heat-slowed charging
Dispatch	Decide and guide	Forecast, Trust, the driver's confidence target	GO or NO-GO, a plan with alternates, the Bingo Line, the point of no return, levers	Neo4j proposes candidate charging chains; the uncertainty simulation with backups decides

The charger graph in Neo4j

One graph links every charger to its evidence and context



Neo4j charger graph · 7 node types, 7 relationships

Each box is a node and each arrow a relationship carrying its numbers. Reliability is inferred from the REPORTED evidence weighted by each source's trust, with networks, sites and hardware supplying starting priors.

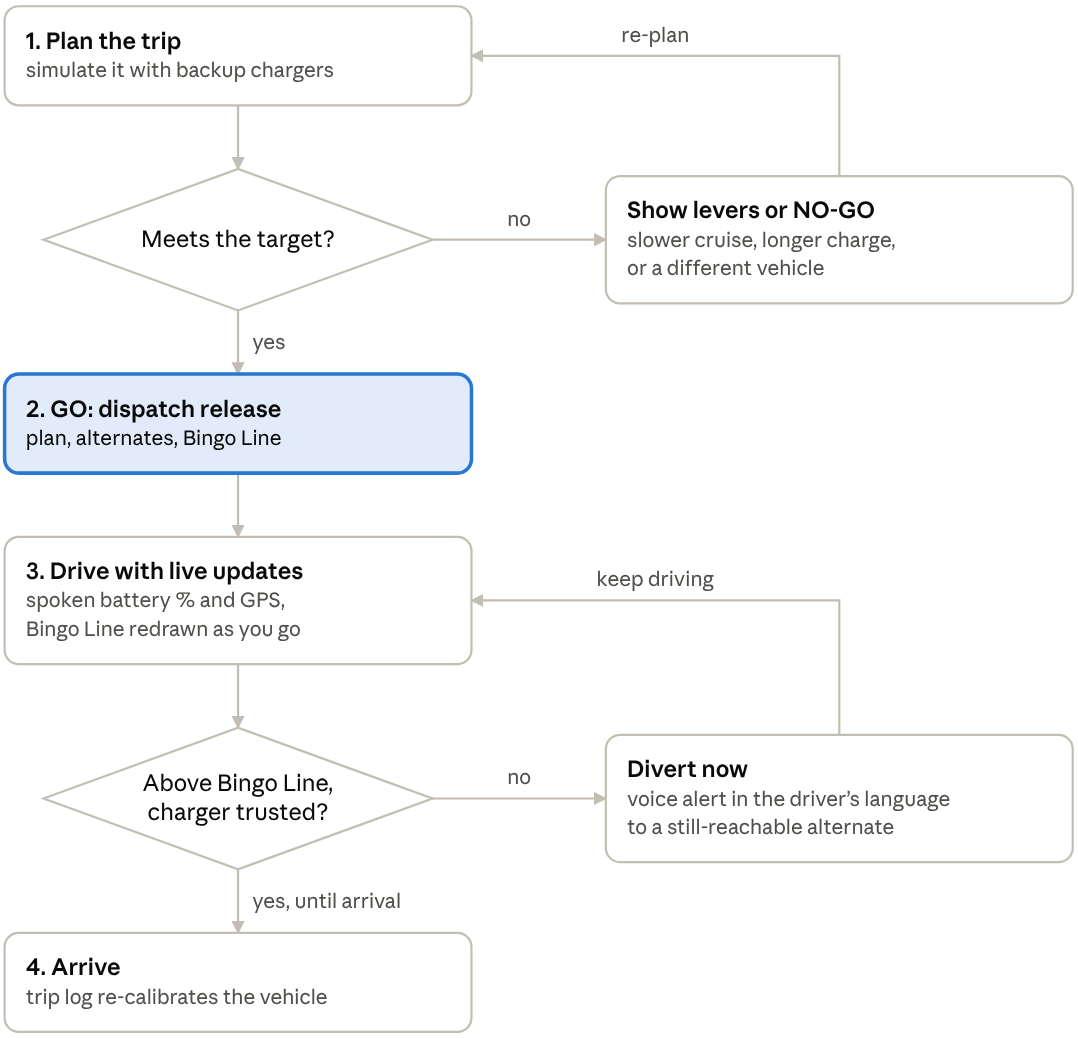
Graph job	How it works	Why judges care
Source trust	Works out the truth from noisy reports and learns how honest each source is	Reliability inferred from reports and usage, as T6.1 asks
Fake-report rings	Community detection finds accounts that report the same chargers in coordinated bursts	A real manipulation defence
"Works, but not for your car"	WORKS_WITH success rates, with similar vehicles filling gaps	Captures car–charger incompatibility
Candidate charging chains	Ranked shortest paths over LEG relationships give several options	The simulation then chooses; the graph never decides alone

Graph job	How it works	Why judges care
Coverage gaps	Road segments with no reliable ALTERNATE	Warnings for drivers, siting insight for networks

One caveat to state openly: reaching the next charger depends on the battery % when leaving the last one, so each LEG is stored per battery band, not as one fixed number.

What happens on a trip

Every trip has a plan B, and the plan updates while you drive



Trip flow · plan, GO, drive, divert, arrive

The plan is never frozen: every spoken battery reading re-runs the forecast, and the divert decision always comes before the point of no return.

Proposed technology stack

Layer	Tool
Models and simulation	Python with NumPy and SciPy
Routes, elevation, weather	An open-source router on OpenStreetMap, open elevation data, Open-Meteo
Charger graph	Neo4j AuraDB Free with Aura Graph Analytics
Voice	Sarvam: Saaras v3 speech-to-text, Bulbul v3 text-to-speech, Sarvam Translate
Backend	Python web API
Interfaces	Web app with a map: fleet console and driver view

6. Data plan

Real Indian data is the scarcest asset and decides how convincing we are, so collection starts tonight, before the model exists.

Source	What it gives us	Who	When
Trip logger (our own web app)	Phone GPS plus spoken or tapped battery % on real Indian trips, on any EV including a scooter	Frontend builds it; every teammate with EV access logs	Built Mon night; logging Tue–Fri
TU Munich BMW i3 dataset	72 real trips with temperature, elevation, battery voltage, current and SoC: a public held-out test set	Modeller	Download Mon night
Driver and fleet interviews	Trip histories, quotes, and one design change we make because of them	Field & Story	Tue–Wed

Source	What it gives us	Who	When
Charger field audit	Real status of 15–20 chargers near Noida: ground truth for the trust engine	Field & Story	Wed (half a day)
Charger directories	Locations and connectors from Open Charge Map and OpenStreetMap, plus a hand-built corridor list from operator apps	Graph & Backend	Tue
Published audits and government counts	Starting reliability priors by network and site type	Graph & Backend	Tue
Open map, elevation and weather data	Routes, gradients, temperature	Modeller	Mon–Tue

Trip logger: minimum spec

- Records GPS every few seconds while a trip is running.
- Driver says or taps the battery % at start, every 10–15 minutes, and at the end.
- Also asks for vehicle model, passengers or load, and whether the AC was on.
- Exports one file per trip; even start and end % alone are useful.

Interview script (10 minutes, e-cab or e-auto drivers, small fleet owners)

1. Can you share your last 10 trips: distance, starting %, ending %? Screenshots from your app are fine.
2. When did you last turn down a trip, or over-charge, because of the battery?
3. Which chargers do you avoid, and how did you learn that?
4. What does a dead charger cost you, in time and in rupees?
5. Would a go/no-go message with a plan B before a booking change what you do? In which language would you want it?

Record exact quotes with permission, and write down one thing in the design that changed because of an answer. The judges explicitly reward that.

Charger audit: what to log for each charger

Field	Example
Location and network	Mall basement, network name
Listed status in its app	"Available"
Actual status on arrival	Working / broken / blocked / locked / no power
Connector types and rated power	CCS2, 60 kW
Access and payment	Open 24 hours, UPI works
Photo and time of visit	Photo file, 3:40 PM

Stay within public, accessible areas, and ask staff before photographing private premises.

7. Validation: five experiments

E1 is the headline result because the judges' first T6.1 criterion is prediction error on trips the model never saw; E2–E5 each prove one other claim.

#	Question it answers	Method	Metric	Compared against
E1	Is it accurate on trips it never saw, after very few trips?	Calibrate on 1, 2, 3, 5 and 10 trips; test on all remaining trips (BMW i3 set, then our Indian trips)	Error in battery-% points; share of outcomes inside the 90% band	Brochure figure; flat "minus 25%" rule; a plain machine-learning model
E2	When Verity says 90%, do 90% of trips arrive?	Generate trips from a harsher "truth simulator" (gusts, driver variability, battery ageing)	Calibration chart: predicted confidence vs actual arrival rate	The same model with segments treated as independent

#	Question it answers	Method	Metric	Compared against
E3	Does charger reliability change decisions?	Plan the same trips with and without the trust engine	How often the charger choice changes; stranding rate; minutes lost	Reliability-blind planner
E4	Do backup plans matter?	Thousands of simulated trips, with charger failures at rates from published audits	Stranding rate	Single fixed shortest-path plan
E5	Does the trust engine find the truth from bad reports?	Feed known failure rates through noisy and partly fake reports	Error in recovered reliability; ring-detection precision	Average star rating

Rules for honest results

- **Hold out test trips before tuning anything,** and never look at them while building.
- **Report where the model is still wrong,** not only where it works. A calibration chart with one honest gap is more convincing than a perfect one.
- **State the compute cost:** time to produce one dispatch plan on a normal laptop.
- **Label simulation as simulation** in every chart title, and keep real-data results visually separate.

A note on E1: the BMW i3 trips are German driving with heater load, not Indian AC load. They prove the calibration method works; our own Indian trips prove it works here.

8. Build schedule and team roles

Four roles work in parallel from Monday night, and we submit on `Oct 31, 2026` by 6 PM IST, leaving six hours of buffer before the portal closes.

Roles

Role	Owns	Experiments	Suggested owner
Modeller	Physics energy model, calibration, uncertainty simulation, Bingo Line, dispatch evaluation	E1, E2, E4	Ansh (ML and predictive-maintenance background)
Graph & Backend	Neo4j charger graph, trust engine, candidate charging chains, API, tests	E3, E5	To assign
Frontend & Voice	Trip logger, fleet console, driver view, Sarvam voice in and out	None	To assign
Field & Story	Interviews, charger audit, logging coordination, deck, demo video	None	To assign

With three people, split Field & Story between Frontend and Graph.

Day by day

Four lanes run in parallel; we submit six hours before the deadline

	Modeller	Graph & Backend	Frontend & Voice	Field & Story
Mon 26 tonight	Get the i3 dataset; physics skeleton	Repo, Neo4j Aura, API skeleton	Trip logger live; team starts logging	Interview script, audit route
Tue 27 add teammates by 11:59 PM	Calibration; first learning curve	Graph schema; import chargers	Console wireframe; Sarvam keys	Driver and fleet interviews
Wed 28	Uncertainty sim; Bingo Line	Trust engine: priors, source trust, usage	Driver view; Bingo Line visual	Charger audit; load into graph
Thu 29	Dispatch, backups, levers; E2, E4	Candidate chains, coverage gaps; E3, E5	Voice in and out; trip replay	Deck draft
Fri 30	E1 on Indian trips; fill abstract numbers	Integration, tests, README	Polish; record the demo video	Deck final; AI-use disclosure
Sat 31 deadline day	**All lanes: buffer and final checks · submit by 6 PM IST, not 11:59 PM**			

Build schedule · 4 lanes, Mon 26 to Sat 31 Oct

Data collection (trip logging, interviews, the charger audit) starts on day one so that real results exist by Friday.

Gates to tick off

- [ ] Trip logger live and shared with the team (Mon night)
- [ ] Any extra teammates added on the portal (by Oct 27, 2026, 11:59 PM)
- [ ] First learning curve on the BMW i3 data (Tue)
- [ ] Charger audit done and loaded into the graph (Wed)
- [ ] All five experiments run at least once (Thu)
- [ ] Abstract numbers filled from real results (Fri)
- [ ] Demo video recorded (Fri)
- [ ] AI and third-party tool use disclosed (Fri)
- [ ] Submitted (by 6 PM on Oct 31, 2026)

9. Demo storyline and deck

The demo shows one fleet booking from decision to arrival in about three minutes, and the deck is designed as an airline "dispatch release" so it looks like nothing else in the room.

The three-minute demo

All numbers below are illustrative until the model produces real ones.

1. **Fleet console:** an outstation booking (Noida to Agra, four passengers, hot afternoon) and three available vehicles.
2. **Vehicle 7** has an aged battery: about 71% confidence, so NO-GO, with the reason shown.
3. **Vehicle 12** needs one stop at a well-trusted charger: about 92%, so GO.
4. **Vehicle 19** relies on a charger with no reports in 40 days. Verity asks the community to verify that one charger before departure.
5. **Driver's phone, trip replay:** the battery drops faster than planned; the estimate updates and a Hindi voice alert says to divert before the point of no return.
6. **Charger failure:** a trusted source reports the planned charger down; the alternate was already reachable, so the trip still succeeds.
7. **Close on the calibration chart:** when Verity said 90%, how many trips actually arrived.

Deck design: the "Dispatch Release"

- Styled like an airline operational flight plan: monospace data tables, GO / NO-GO stamps, amber on charcoal.
- Each slide headed like a dispatch form field, e.g. "SECTION 4: RESERVE POLICY".
- A route strip with its Bingo Line runs along the bottom of the key slides.
- No stock icons and no template layouts; every visual is a real chart, screenshot or diagram from our build.

Slide plan

#	Slide	What it proves
1	Cover: a dispatch release for one booking, stamped GO at about 90%	The concept in one glance

#	Slide	What it proves
2	One number lies: claimed vs tested range, the flat-discount rule, charger audit figures	The problem is real and measured
3	How airlines do it: destination, alternate, reserve, bingo fuel	The core insight
4	What the driver sees: the Bingo Line	Honest uncertainty in the interface
5	Four engines, mapped to every bullet of T6.1	Problem and track alignment
6	E1: learning curve and error on unseen trips	Accuracy and calibration from sparse data
7	E2: when we say 90%, it is 90%	Honest confidence
8	Chargers that actually work: the graph, E3, E5 and our audit	Reliability changes decisions; Neo4j track
9	Shortest path vs honest planning: E4	Stochastic planning, the layer beneath
10	Fleet console and driver voice	Functionality; Sarvam track
11	Who pays: fleets now, then used-EV range certificates, OEMs and charging networks	Business case and data moat
12	What we don't know yet, and the finale plan	Critical thinking
Appendix	AI and tool disclosure, assumptions, data sources	Transparency

10. Risks and judge questions

The biggest risk is that too much of our evidence is simulated, so real trips, interviews and the charger audit are scheduled first, not last.

Risks and fallbacks

Risk	Likelihood	Impact	Fallback
Few or no real Indian trips logged	Medium	High: weakens E1	Use the BMW i3 set for the method; log even short scooter trips; use interviewed drivers' trip histories
Charger reports are mostly simulated	High	Medium	Ground the simulation in published audit rates and our own audit; prove the engine recovers known truth (E5)
Scope too large for five days	High	High	Must-ship list below; cut the stretch items first
Neo4j or Sarvam setup issues	Low	Medium	Neo4j AuraDB Free or a local install; text alerts if voice fails
A teammate drops out	Medium	Medium	Roles merge as described in section 8
Last-minute submission problems	Medium	High	Submit by 6 PM on 31 Oct, not 11:59 PM

Must ship vs stretch

- **Must ship:** physics model and calibration with the learning curve; uncertainty simulation and Bingo Line; Neo4j charger graph with priors, source trust and usage traces; dispatch evaluation with backups; fleet console and driver view with voice alerts; experiments E1–E4.
- **Stretch:** full fake-report ring detection (E5), car–charger compatibility model, community verification requests, reading the battery % from a dashboard photo.

Hard questions to rehearse

Judge asks	Our answer
"Your charger data is simulated."	Locations are real, priors come from published audits, we audited chargers ourselves, and E5 shows the engine recovers known truth. Real reports replace simulation as they arrive.
"Why not just use machine learning?"	There is no Indian training data to learn from. E1 shows physics plus priors beats a plain ML model at low trip counts, and every number can be explained.
"A BMW i3 in Germany isn't India."	It proves the calibration method. Our own Indian trips form the second test set, and AC load enters the same climate-load term as heating.
"How do you know 90% means 90%?"	The calibration chart from E2, including where it is still off.
"Isn't this just A Better Routeplanner?"	ABRP uses a fixed arrival buffer, its users have asked for alternate-charger logic, it has no source-trust model, and it is not built for Indian conditions.
"Who pays, and why now?"	Fleets: a dead vehicle costs a whole shift. Used-EV buyers: India has no trusted battery-health signal. Fleet data then improves the model for everyone.
"What if drivers don't report battery %?"	Start and end readings alone still work; voice input makes mid-trip readings hands-free; fleet telematics can feed it directly later.

11. Beyond the prototype

At the finale we want to arrive with a pilot story: one real fleet operator whose vehicles Verity has already calibrated.

At the finale (Nov 21, 2026 to 22 Nov, IIIT Delhi)

- Calibrate on a real fleet's vehicles, using an operator we met during the interviews.
- Turn the battery-capacity estimate into a used-EV **real-range certificate**.
- Add the car–charger compatibility model and full fake-report detection if they were cut before 31 Oct.
- Run one live trip with the driver view and voice alerts.

Who pays

Buyer	Their pain	What they pay for
Small EV fleets (e-cabs, cargo three-wheelers)	A dead vehicle in the wrong place costs a whole shift	Go/no-go dispatch per booking; fewer strandings and wasted charging stops
Used-EV buyers, sellers and dealers	No trusted battery-health signal, so used EVs are priced at the worst case	A real-range certificate built from actual trips
Charging networks	No independent view of which of their chargers fail	Reliability and coverage-gap reports
Vehicle makers	Range claims that customers distrust	Real-world energy data by model and region

Why it gets harder to copy over time

Every trip from every fleet vehicle sharpens the shared priors, so new vehicles become accurate faster. A newcomer starts with no trips, which is the data loop the problem statement itself names as the long-term advantage.

12. Glossary

Plain-language definitions of every technical term used in this doc.

Term	Meaning
SoC (state of charge)	The battery %, as shown on the dashboard

Term	Meaning
Usable capacity	How much energy the battery really holds today; it shrinks as a battery ages
Claimed range	The range from the official lab test (in India, the MIDC cycle)
Calibration	Adjusting the model's settings so its predictions match one specific vehicle
Physics energy model	Energy computed from forces: rolling resistance, air drag, climbing, accelerating, plus AC load
Prior	What the model assumes before seeing a vehicle's own trips, e.g. the typical efficiency of that car model
Bayesian update	A method that blends the prior with each new trip, so estimates sharpen as data arrives
Fleet pooling	Vehicles of the same model share what they learn, so a new vehicle is useful from its first trip
Monte Carlo simulation	Running a trip thousands of times with random variations to see the spread of outcomes
Shared (correlated) uncertainty	Errors that persist through a whole trip, like a heavy load; ignoring them makes long trips look falsely safe
Confidence band	The range of likely outcomes, e.g. "arrive with 8–19%"
Calibration chart	Plots predicted confidence against how often trips actually arrived; a well-calibrated model sits on the diagonal
Held-out trips	Trips kept away from the model while building it and used only for testing
Dispatch release	The airline document that clears a flight to depart with its fuel plan, alternates and reserves; Verity's output for a trip

Term	Meaning
Alternate	A backup charger reachable if the planned one fails, like an airline's alternate airport
Bingo Line	The battery % at each point of the route below which the driver must divert; from the pilots' term "bingo fuel"
Point of no return	The last point on the route from which an alternate is still reachable
Recourse	The ability to change plan mid-trip; Verity's probabilities include it
Chance-constrained planning	Choosing the fastest plan that still meets a required success probability, e.g. 90%
Stochastic planning	Planning when the outcome is uncertain, as opposed to a fixed shortest route
Charger trust engine	The part of Verity that estimates whether each charger will work for your car right now
Source trust (Dawid–Skene model)	A method that works out the truth from many noisy reports while learning how reliable each reporter is
Usage traces	Indirect evidence, e.g. drivers who arrive at a charger and leave within minutes without charging
Fake-report ring	A group of accounts posting coordinated false reports, e.g. to harm a rival network
Neo4j	A graph database that stores items as nodes and the links between them as relationships
Graph	Items (chargers, networks, vehicles, reporters) connected by relationships (reported, runs, is compatible with)
Coverage gap	A stretch of road with no reliable alternate charger within reach
Sarvam	An Indian AI company whose speech-to-text and text-to-speech models cover Indian languages

Term	Meaning
OEM	Original equipment manufacturer: the vehicle maker
CPO	Charge point operator: the company that runs a charging network
ABRP	A Better Routeplanner, a widely used global EV trip planner